

4(a)(i)	distance from a (reference) point in a given direction	B1
4(a)(ii)	line is not straight or gradient is not constant	B1
4(b)(i)	0.85–0.90 cm	A1
4(b)(ii)	$a = -(2\pi f)^2 x$	C1
	e.g. $1.2 = 4\pi^2 \times f^2 \times (0.90 \times 10^{-2})$	C1
	$f = 1.8$ Hz	A1
4(c)	complete circle/ellipse enclosing the origin	B1
	closed shape passing through $(0, \pm v_0)$ and $(\pm x_0, 0)$	B1